

On the Stability of Microfluidic Drug Carriers

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Abstract

Work in biomedical research has long relied on assumptions we test directly here. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The reduced model retains most of the explanatory power at a fraction of the computational cost. Calibration drift over the trial period remained below the threshold set out in the protocol.

1. Introduction

Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting.

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Calibration drift over the trial period remained below the threshold set out in the protocol. The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. Longer observation windows narrow the confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

2. Related Work

Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Sensitivity to the smoothing window proved modest across the range we examined. Calibration drift over the trial period remained below the threshold set out in the protocol. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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3. Materials and Methods

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Calibration drift over the trial period remained below the threshold set out in the protocol. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. Neither of the alternative formulations improved on the simple additive model in our setting. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

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4. Experimental Setup

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5. Results

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6. Discussion

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7. Limitations

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8. Conclusion

Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Neither of the alternative formulations improved on the simple additive model in our setting. Residual error grows approximately linearly with the number of coupled parameters.

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